Pablo Naval Baudín, MD

Consultant Neuroradiologist · Clinical AI and Imaging Biomarkers

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Consultant neuroradiologist and clinical researcher specialising in multiple sclerosis, neuro-oncology, X-linked adrenoleukodystrophy and presurgical functional mapping. Clinical Lead of the Bellvitge Radiomics Research and Transformation Team and Coordinator of SERAM's Artificial Intelligence Commission. Leads translational projects spanning imaging biomarkers, clinical AI implementation and radiology workflow redesign.

CURRENT APPOINTMENTS

- **May 2018–Present · Consultant Neuroradiologist**
  Institut de Diagnòstic per la Imatge (IDI), Hospital Universitari de Bellvitge · Barcelona, Spain
- **June 2020–Present · Consultant Neuroradiologist**
  Hospital Quirónsalud Barcelona · Barcelona, Spain
- **August 2022–Present · Clinical Lead, Radiomics Research and Transformation Team**
  Department of Radiology, Hospital Universitari de Bellvitge · Barcelona, Spain
- **February 2024–Present · Clinical Researcher**
  Bellvitge Translational Imaging Biomarkers Group, Institut d'Investigació Biomèdica de Bellvitge (IDIBELL)
- **November 2025–Present · Honorary Research Fellow**
  Department of Neuroinflammation, Queen Square Institute of Neurology, University College London (UCL)

Previous Appointment

- **September 2019–March 2022 · Neuroradiologist**
  Telemedicine Clinic, UK Reporting Team

LEADERSHIP AND INSTITUTIONAL ROLES

- **July 2026–Present · Coordinator, Artificial Intelligence Commission**
  Spanish Society of Medical Radiology (SERAM)
- **2025–2026 · Leadership roles, 38th SERAM National Congress**
  Innovation and Information Technology Lead, Scientific Committee; Head of the Biotechnology and Artificial Intelligence Committee; Member, Oncologic Imaging Section · Valencia, Spain
- **2025–Present · Member, Young Committee**
  Spanish Society of Neuroradiology (SENR)
- **2024 and 2025 · Course Lecturer and Project Mentor**
  Master's Degree in Artificial Intelligence in Radiology, Universitat de Barcelona
- **2018, 2019 and 2021–2025 · Lecturer and Scientific Committee Member**
  International Course on Brain Mapping, Hospital Universitari de Bellvitge

SELECTED LECTURES, PANELS AND WORKSHOPS

- **Invited lecturer, "AI-driven redesign of the MRI pathway in multiple sclerosis"**
  Simposio Internacional MRI 2026, Clínica del Country and Clínica La Colina · Bogotá (online) · 5 November 2026
- **Invited lecturer, "Cerrando la brecha: Traslación clínica de la IA y procesado de imagen médica"**
  II Jornada Anual IABiomed · Barcelona, Spain · 30 October 2026 · Official programme
- **Faculty, "Presurgical fMRI Hands-on Workshop"**
  4th ESNR Advanced Course in Diagnostic Neuroradiology on Glioma Imaging · Madrid, Spain · 11–13 June 2026
- **Invited lecturer, "Advanced Diffusion: What Are We Missing?"**
  38th SERAM National Congress · Valencia, Spain · 21 May 2026 · Also moderator of seven scientific sessions
- **Panellist, "Innovation and Future Vision"**
  Radiology and Digital Health: Current Reality · Girona, Spain · 9 April 2026
- **Invited webinar lecturer, "Advanced Diffusion in Brain Tumors"**
  Radiomics and Advanced Neuroimaging in Neuro-oncology · Online · 26 June 2025
- **Invited lecturer, "The Role of Radiomics in Pituitary and Neuroendocrine Tumours"**
  XVII Annual Scientific Meeting of the Neuroendocrinology Knowledge Area · Madrid, Spain · 31 January 2025
- **Lecturer, "Programming Languages and Technical Components"**
  Introduction to Artificial Intelligence in Radiology · Barcelona, Spain · 9 May 2024
- **Faculty, "Hands-on fMRI Workshop"**
  3rd ESNR Advanced Course in Diagnostic Neuroradiology on Glioma Imaging · Madrid, Spain · 18 November 2022
- **Invited panellist, "How to Train Radiologists and Related Professionals in AI"**
  European Congress of Radiology 2022 · Vienna, Austria · 14 July 2022

AWARDS AND RECOGNITION

- **2026 · Winner, Category II ("Presente digital")** · IESE–Novartis Award for Operational Excellence in Healthcare Management, for ImaginEM: AI-driven redesign of the MRI pathway in multiple sclerosis · Project Coordinator
- **2026 · Certificate of Merit** · "T1-dark rim: Can it predict secondary progressive multiple sclerosis?" · 38th SERAM National Congress
- **2024 · Certificate of Merit** · "Quantifiable deeply hypointense foci on 3D T1 TFE are characteristic of paramagnetic-rim lesions in multiple sclerosis" · 37th SERAM National Congress
- **2021 · Magna Cum Laude** · "Post-contrast SWI in Multiple Sclerosis Follow-up MRI" · SERAM National Congress
- **2020 · Certificate of Merit** · "MRI and PET/CT Findings in Creutzfeld-Jakob Disease: Differentiating Between Different Forms" · American Society of Neuroradiology (ASNR) Annual Meeting
- **2019 · Best Scientific Presentation Award** · "Assessment of the Effect of Temporal and Insular Lesions on the Inferior Fronto-Occipital Fascicle Using DTI and Tractography" · 48th Annual Meeting, Spanish Society of Neuroradiology (SENR)
- **2019 · Best Scientific Presentation Award** · "Differentiation Between Primary CNS Lymphoma and Anaplastic Astrocytoma Using a Novel T2* Perfusion Curve Analysis Method" · Catalan Society of Radiology (ACRAM) Annual Meeting
- **2018 · Certificate of Merit** · "Exploring Eloquent Frontal Areas by fMRI & DTI in Neurosurgical Patients" · Radiological Society of North America (RSNA) Annual Meeting
- **2018 · Second Best Scientific Presentation Award** · "Differentiation between Primary CNS Lymphoma and Anaplastic Astrocytoma Using a Novel Method for T2* Perfusion Curve Assessment" · 47th Annual Meeting, Spanish Society of Neuroradiology (SENR)
- **2018 · AIRP–SERAM Scholarship** · Scholarship to attend the four-week AIRP Radiologic Pathology Correlation Course · Maryland, USA
- **2017 · Cum Laude** · "Imaging of Skull Vault Tumors in Adults: Diagnosing in No-man's Land" · Radiological Society of North America (RSNA) Annual Meeting
- **2017 · ECR Invest in the Youth Scholarship** · European Society of Radiology · European Congress of Radiology 2017 · Vienna, Austria
- **2016 · Third Best Scientific Presentation Award** · "Positive Airway Pressure-enhanced CT to Improve Virtual Bronchoscopic Navigation" · AABIP–CHEST Annual Meeting

SELECTED PROJECTS

- **2026 · IA-ICTUS: Independent Validation of the Siemens-Methinks Stroke Suite**
  Internal IDI project · Project Coordinator; Reference-Standard and Ground-Truth Lead · Validation for AQuAS and Pla Director before planned regional deployment in CatSalut
  Coordinated the multicentre study and neuroradiologist reference-standard workflow, including blinded double reading and adjudication, for 180 CT cases and a 65-patient CTP/MRI subcohort.
- **2024–2026 · MSPredict: MS Prognosis through Advanced Imaging Analysis**
  Project Lead · CaixaImpulse Innovation in Health 2024, "la Caixa" Foundation · Ref. CI24-10486 · €51,000
  Development of a clinical decision-support tool to predict disability progression in multiple sclerosis through advanced machine learning and multimodal data integration.
- **2022–2025 · Multi-omics Medicine to Understand Adrenomyeloneuropathy**
  Collaborator · Phase IIb/III randomised trial of dimethyl fumarate · Instituto de Salud Carlos III · Ref. ICI21/00085 · €898,700 · PI: Aurora Pujol Onofre
  Contributes neuroimaging analysis, including diffusion tractography and MR spectroscopy, to this multicentre placebo-controlled trial.
- **2023–2025 · HIPOMEM: Role of the Hippocampus in Memory Processes in Temporal Lobe Epilepsy**
  Collaborator · Ministry of Science and Innovation · Ref. CNS2022-136160 · €199,408 · PI: Estela Cámara Mancha
  Study of hippocampal memory function to support surgical planning in mesial temporal lobe epilepsy.

RESEARCH FOCUS

- **Multiple sclerosis** · Imaging biomarkers for disease diagnosis, prognosis and disability progression
- **X-linked adrenoleukodystrophy** · Prognostic imaging biomarkers and quantitative neuroimaging
- **Brain tumour surgical planning** · Functional MRI and diffusion tractography mapping

EDUCATION AND SPECIALIST TRAINING

- **September 2022–Present · PhD Candidate in Translational Medicine (Neurosciences)**
  Faculty of Medicine, Universitat de Barcelona · Research focus: predictive neuroimaging biomarkers in multiple sclerosis
- **May 2014–May 2018 · Radiology Residency and Specialist Training**
  Hospital Universitari de Bellvitge · Barcelona, Spain
- **September–December 2017 · Visiting Rotation in Neuroradiology**
  University of North Carolina at Chapel Hill · North Carolina, USA
- **September–December 2012 · Medical Elective**
  Centre Hospitalier Universitaire Vaudois · Lausanne, Switzerland

- **September 2007–June 2013 · Degree in Medicine**
  Universitat Autònoma de Barcelona, Vall d'Hebron Teaching Unit · Barcelona, Spain · Highest honours in Biostatistics and Advanced Histology; honours in General Pathology, Pharmacology, Clinical Pharmacology, Emergency Medicine, History of Medicine and Clinical Internship

ACADEMIC SUPERVISION

Radiology Residency Tutor · Hospital Universitari de Bellvitge (2021–Present). Supervised eight radiology residents.

- **Current cohort · 2024–2028**
  Dr Carlos Jaber, Dr Enric Vidal, Dr Laura López and Dr Yuzin Zhou
- **Completed cohort · 2020–2024**
  Laura Chavarriaga, Dr Ilyan Mezinskiy, Dr Enric Munté and Dr Albert Castillo

Final-year projects and engineering placements

- **Mar González Montesinos · BSc Biomedical Engineering, UPC · 2024**
  Bachelor's thesis: Automated Loes score assessment on brain MRI in adrenomyeloneuropathy
- **Héctor Vadillo Valero · BSc Biomedical Engineering, UPC · 2023**
  Bachelor's thesis: AI tool for morphometric follow-up of brain tumours
- **Iria Varona Cillero · BSc Biomedical Engineering, UPC · 2024**
  Placement: DSC perfusion MRI in neuro-oncology
- **Nicolas Auger · Engineering, IMT Atlantique · 2024**
  Placement: Optimisation of QSM processing pipelines

SCIENTIFIC PROFILE

- **33 MEDLINE-indexed publications**
  h-index: 12 (Web of Science), 12 (Scopus) and 14 (Google Scholar)
- **Peer reviewer for five Q1 journals**
  European Radiology, Academic Radiology, Journal of Magnetic Resonance Imaging, Clinical Neuroradiology and AJNR

BOOK CHAPTER

- **Naval-Baudin P**, Romera-Lobato N. **Procesamiento de imágenes médicas con IA**. In: Cabero Roura L, Guzmán López A (eds). "IA. Medicina e ingeniería. Algoritmos que salvan vidas". Universidad Autónoma de Nuevo León; 2025 (in production).

ADVANCED TRAINING AND CERTIFICATIONS

- **Advancing Clinical Practice Through AI Innovation** · Radiological Society of North America · 2025
- **Artificial Intelligence 4 Imaging** · Maastricht University, Netherlands · 2023
- **Deep Learning with PyTorch for Medical Image Analysis** · Udemy · 2023
- **FSL Course** · University of Oxford, UK · 2022
- **Advanced Diagnostic Course in Neuroradiology: Imaging in Epilepsy** · European Society of Neuroradiology · Barcelona, Spain · 2022
- **European Course in Neuroradiology (ECNR), Modules 1–4** · European Society of Neuroradiology (ESNR) · 2019–2021
- **AIRP Radiologic Pathology Course** · American Institute for Radiologic Pathology · 2018

SELECTED MEDIA ENGAGEMENT

- **2025 · Featured project lead, "Can AI Predict the Course of Multiple Sclerosis?"** · La Vanguardia, Big Vang · 10 March 2025
- **2024 · Television interview on radiomics and clinical AI implementation** · TV L'Hospitalet, L'Entrevista de l'Informatiu · 31 October 2024
- **2024 · National television and radio coverage of AI implementation in multiple sclerosis imaging** · Telecinco Informativos · 9 September 2024; Onda Cero Catalunya · 4 September 2024
- **2023 · Media coverage of clinical AI, radiomics and multiple-sclerosis imaging biomarkers** · TV3 Telenotícies and La Vanguardia · 20 February; Catalunya Ràdio, Revolució 4.0 · 4 March; La Vanguardia · 16 August; Redacción Médica · 11 September

LANGUAGES

Spanish and Catalan · Native / bilingual • **English** · C2 · IELTS 8.5/9; TOEFL iBT 117/120 • **French** · C1

Annex: PubMed-Indexed Publications

33 publications · Updated July 2026 · Author name highlighted in bold

2026

1. **Naval-Baudin P**, Rovira A. Implicaciones Radiológicas de Los Criterios Revisados McDonald de 2024 Para La Esclerosis Múltiple. Radiología, 2026, 501811
2. **Naval-Baudin, P**, González-Montesinos M. Time to Tap into Clinical MRI for Brain Volumetry Research? European Radiology, 2026, 2–4

2025

3. **Naval-Baudin P**, Bermúdez Bravo WF, Pineda-Borja VI, et al. 3D T1 Turbo Spin Echo Improves Detection of Gadolinium-Enhancing Multiple-Sclerosis Lesions. *Insights Imaging* 2025.
4. **Naval-Baudin P**, Pérez-Alfonso K, Castillo-Pinar A, et al. Post-contrast Susceptibility Weighted Imaging in Multiple Sclerosis MRI Improves the Detection of Enhancing Lesions. Clin Neuroradiol 2025;35:533–9.
5. Tena-Cucala R, **Naval-Baudin P**, Martínez-Yélamos A, et al. The 2024 McDonald criteria show comparable sensitivity to previous criteria for late-onset multiple sclerosis. *Mult Scler Relat Disord* 2025;103.
6. Pareto D, **Naval-Baudin P**, Pons-Escoda A, et al. Image analysis research in neuroradiology: bridging clinical and technical domains. *Neuroradiology* 2025;67:1649–66.
7. **Naval-Baudin P**, Pons-Escoda A. Reconsidering gadolinium in long-term glioma follow-up: is contrast always and unquestionably necessary? *Eur Radiol* 2025;35:5203–5.

2024

8. **Naval-Baudin P**, Pons-Escoda A, Camins À, et al. Deeply 3D-T1-TFE hypointense voxels are characteristic of phase-rim lesions in multiple sclerosis. *Eur Radiol* 2024;34:1337–45.
9. **Naval-Baudin P**, Pons-Escoda A, Castillo-Pinar A, et al. The T1-dark-rim: A novel imaging sign for detecting smoldering inflammation in multiple sclerosis. *Eur J Radiol* 2024;173:111358.
10. **Naval-Baudin P**, Arroyo-Pereiro P, Majós C. The pressing need for imaging biomarkers of disability progression in multiple sclerosis. *Eur Radiol* 2024;34:3823–5.
11. Garcia-Ruiz A, Pons-Escoda A, Grussu F, et al. An accessible deep learning tool for voxel-wise classification of brain malignancies from perfusion MRI. *Cell reports Med* 2024;5:101464.
12. Pons-Escoda A, Garcia-Ruiz A, **Naval-Baudin P**, et al. Differentiating IDH-mutant astrocytomas and 1p19q-codeleted oligodendrogliomas using DSC-PWI: high performance through cerebral blood volume and percentage of signal recovery percentiles. Eur Radiol. 2024;34:5320-5330.
13. Majós C, Pons-Escoda A, **Naval-Baudin P**, et al. Proton MR spectroscopy shows improved performance to segregate high-grade astrocytoma subgroups when defined with the new 2021 World Health Organization classification of central nervous system tumors. Eur Radiol. 2024;34:2174-2182.
14. Pons-Escoda A, **Naval-Baudin P**, Viveros M, et al. DSC-PWI presurgical differentiation of grade 4 astrocytoma and glioblastoma in young adults: rCBV percentile analysis across enhancing and non-enhancing regions. *Neuroradiology* 2024;66:1267–77.
15. **Naval-Baudin P**, Pons-Escoda A. "Pushing the Envelope": advanced imaging-data-analysis meets NODDI to differentiate glioblastoma and brain metastasis. *Eur Radiol* 2024;34:6614–5.

2023

16. Pons-Escoda A, Garcia-Ruiz A, Garcia-Hidalgo C, et al. MR dynamic-susceptibility-contrast perfusion metrics in the presurgical discrimination of adult solitary intra-axial cerebellar tumors. Eur Radiol 2023;33:9120–9.
17. Pons-Escoda A, **Naval-Baudin P**, Velasco R, et al. Imaging of Lymphomas Involving the CNS: An Update-Review of the Full Spectrum of Disease with an Emphasis on the World Health Organization Classifications of CNS Tumors 2021 and Hematolymphoid Tumors 2022. *Am J Neuroradiol* 2023;44:358–66.
18. Tena-Cucala R, Sala-Padró J, Jaraba S, et al. Eating-induced seizures: A semiological sign of the right temporal pole. *Epileptic Disord* 2023;25:480–91.

2022

19. Pons-Escoda A, Garcia-Ruiz A, **Naval-Baudin P**, et al. Voxel-level analysis of normalized DSC-PWI time-intensity curves: a potential generalizable approach and its proof of concept in discriminating glioblastoma and metastasis. *Eur Radiol* 2022.
20. Camins À, Naval-Baudin P, Majós C, et al. Inferior fronto-occipital fascicle displacement in temporoinsular gliomas using diffusion tensor imaging. *J Neuroimaging* 2022;32:638–46.
21. Pons-Escoda A, García-Ruíz A, **Naval-Baudin P**, et al. Diffuse Large B-Cell Epstein-Barr Virus-Positive Primary CNS Lymphoma in Non-AIDS Patients: High Diagnostic Accuracy of DSC Perfusion Metrics. *AJNR Am J Neuroradiol* 2022;43:1567–74.
22. Gomez-Andres A, Cunillera T, Rico I, et al. The role of the anterior insular cortex in self-monitoring: A novel study protocol with electrical stimulation mapping and functional magnetic resonance imaging. *Cortex* 2022;157:231–44.

2021

23. **Naval-Baudin P**, Rodriguez Caamaño I, Rubio-Maicas C, et al. COVID-19 and Ischemic Stroke: Clinical and Neuroimaging Findings. *J Neuroimaging* 2021;31:62–6.
24. Garcia-Ruiz A*, **Naval-Baudin P***, Ligero M, et al. Precise enhancement quantification in post-operative MRI as an indicator of residual tumor impact is associated with survival in patients with glioblastoma. *Sci Rep* 2021;11:695. *Joint first co-authors
25. Alemany M, Domènech M, Argyriou AA, et al. Perilesional edema in brain metastases as predictive factor of response to systemic therapy in non-small cell lung cancer patients: a preliminary study. *Ann Transl Med* 2021;9:648–648.
26. Gifreu A, Falip M, Sala-Padró J, et al. Risk of Developing Epilepsy after Autoimmune Encephalitis. *Brain Sci* 2021;11:1182.

2020

27. Pons Escoda A, **Naval Baudin P**, Mora P, et al. Imaging of skull vault tumors in adults. *Insights Imaging* 2020;11:23.
28. Pons-Escoda A, **Naval-Baudín P**, Majós C, et al. Neurologic Involvement in COVID-19: Cause or Coincidence? A Neuroimaging Perspective. *Am J Neuroradiol* 2020;41:1365–9.
29. Pons-Escoda A, Garcia-Ruiz A, **Naval-Baudin P**, et al. Presurgical Identification of Primary Central Nervous System Lymphoma with Normalized Time-Intensity Curve: A Pilot Study of a New Method to Analyze DSC-PWI. *Am J Neuroradiol* 2020;41:1816–24.
30. del Cerro I, Martínez-Zalacaín I, Guinea-Izquierdo A, et al. Locus coeruleus connectivity alterations in late-life major depressive disorder during a visual oddball task. NeuroImage Clin 2020;28:102482.

2018

31. **Naval Baudin P**, Pons Escoda A, Cos Domingo M, et al. Invasive Sinonasal Lesions: From the Nasal Fossa and Paranasal Sinuses to the Endocranium. *Curr Probl Diagn Radiol* 2018;47:168–78.

2017

32. **Naval-Baudin P**, Carreño García E, Sanchez Marquez A, et al. Multicystic seminal vesicle with ipsilateral renal agenesis: two cases of Zinner syndrome. *Scand J Urol* 2017;51:81–4.
33. **Naval Baudin P**, Martínez Fernández T, Mora P, et al. Posterior reversible encephalopathy syndrome (PRES) with obstructive hydrocephalus. *Diagn Interv Imaging* 2017;98:745–6.